

EXERCISE 0.1: Let S be a set and $\phi : S \rightarrow S$ be a self mapping of S .

- a) If S is finite, then ϕ is bijective if and only if $\text{Per}(\phi, S) = S$
- b) In general, if $\text{Per}(\phi, S) = S$, then ϕ is bijective
- c) Provide an example of an infinite set S and a self map ϕ of S such that ϕ is bijective but $\text{Per}(\phi, S) \neq S$.
- d) If ϕ is injective, then $\text{PrePer}(\phi, S) = \text{Per}(\phi, S)$

SOLUTION:

- a) A bijection ϕ on a finite set S is a permutation of the elements of S , hence can be written as a product of cycles of elements in S . Any element $\alpha \in S$ is contained in one of these cycles, and thus is ϕ -periodic with period equal to the length of the cycle that contains it.

Conversely, suppose that $\text{Per}(\phi, S) = S$ so that for any $\alpha \in S$ there exists some $m \in \mathbb{N}$ such that $\phi^m(\alpha) = \alpha$. We can see that

$$\alpha = \phi^m(\alpha) = \phi(\phi^{m-1}(\alpha)),$$

hence ϕ maps $\beta = \phi^{m-1}(\alpha)$ to α , proving that ϕ is surjective.

Next, fix $\alpha, \beta \in S$ such that $\phi(\alpha) = \phi(\beta)$, and choose integers $m, n \in \mathbb{N}$ such that $\alpha = \phi^m(\alpha)$ and $\beta = \phi^n(\beta)$. Fixing $k \in \mathbb{N}$, we have

$$\phi^k(\alpha) = \phi^{k-1}(\phi(\alpha)) = \phi^{k-1}(\phi(\beta)) = \phi^k(\beta),$$

hence $\phi^k(\alpha) = \phi^k(\beta)$ for all k . Using this fact, we find that

$$\begin{aligned} \alpha &= \phi^m(\alpha) = \phi^m(\beta) = \phi^m(\phi^n(\beta)) = \phi^n(\phi^m(\beta)) \\ &= \phi^n(\phi^m(\alpha)) = \phi^n(\alpha) = \phi^n(\beta) = \beta, \end{aligned}$$

showing that ϕ is injective, and thus bijective. ■

- b) Since the second part of the previous doesn't rely on the finiteness of S , we are finished. ■
- c) Let $S = \mathbb{Z}$ and define $\phi(n) = n + 1$. Obviously ϕ is bijective, but for any $n \in \mathbb{Z}$, the orbit $\mathcal{O}_\phi(n) = \{k \in \mathbb{Z} : k \geq n\}$ is an infinite set, hence $\text{Per}(\phi, \mathbb{Z}) = \emptyset \neq \mathbb{Z}$.
- d) Assume ϕ is injective and let $\alpha \in \text{PrePer}(\phi, S)$. Also choose $m, n \in \mathbb{N}$ such that $m < n$ and $\phi^m(\alpha) = \phi^n(\alpha)$. We have that

$$\phi^m(\alpha) = \phi(\phi^{m-1}(\alpha)) = \phi^n(\alpha) = \phi(\phi^{n-1}(\alpha)),$$

thus $\phi^{m-1}(\alpha) = \phi^{n-1}(\alpha)$ since ϕ is injective. Performing induction on n gives us $\alpha = \phi^{n-m}(\alpha)$, proving that $\alpha \in \text{Per}(\phi, S)$. Since periodic points are trivially preperiodic, we have shown that $\text{PrePer}(\phi, S) = \text{Per}(\phi, S)$. ■

EXERCISE 0.2: Let S be a set, and let ϕ and ψ be self maps of S . Further suppose that ϕ and ψ commute, i.e. $\phi \circ \psi = \psi \circ \phi$.

- a) We have $\psi(\text{PrePer}(\phi)) \subseteq \text{PrePer}(\phi)$
- b) If ψ is surjective and finite-to-one (i.e. if the preimage of any point in S under ψ is a finite set), then $\psi(\text{PrePer}(\phi)) = \text{PrePer}(\phi)$
- c) A point $\alpha \in S$ is called an **isolated preperiodic point** if there exist $m, n \in \mathbb{N}$ such that $m < n$, $\phi^m(\alpha) = \phi^n(\alpha)$, and the set

$$\{\beta \in S : \phi^m(\beta) = \phi^n(\beta)\}$$

is finite.

If every preperiodic point of ϕ is isolated, then $\text{PrePer}(\phi) \subseteq \text{PrePer}(\psi)$.

SOLUTION: a

EXERCISE 0.3: Let $\phi \in \mathbb{Z}[X]$ with $\phi(X) = X^d + a$, and let p be a prime. We have that $\text{Per}(\phi, \mathbb{F}_p) = \mathbb{F}_p$ if and only if $\gcd(d, p-1) = 1$

SOLUTION:

EXERCISE 0.4: Let G be a group and $\phi : G \rightarrow G$ a group homomorphism.

- a) $\text{Per}(\phi, G)$ is a subgroup of G .
- b) $\text{PrePer}(\phi, G)$ is not necessarily a subgroup of G .

SOLUTION:

- a) Let $\alpha, \beta \in \text{Per}(\phi, G)$ with periods m and n , respectively. Induction on k easily shows that $\phi^k(\alpha\beta) = \phi^k(\alpha)\phi^k(\beta)$, hence

$$\phi^{mn}(\alpha\beta) = \phi^{mn}(\alpha)\phi^{mn}(\beta).$$

Since iterating a periodic point by a multiple of its period fixes the point, we conclude that $\phi^{mn}(\alpha)\phi^{mn}(\beta) = \alpha\beta$ and that $\alpha\beta \in \text{Per}(\phi, G)$. Additionally,

$$\phi^m(\alpha^{-1}) = (\phi^m(\alpha))^{-1} = \alpha^{-1},$$

hence $\alpha^{-1} \in \text{Per}(\phi, G)$, proving that $\text{Per}(\phi, G)$ is a subgroup of G . ■

- b) a

EXERCISE 0.5: Let G be a topological group and $\phi : G \rightarrow G$ a continuous homomorphism on G . Compute the following topological closures: